Chapter 171

Setting Up Tracks, Busses, and Patching

One of the first things you need to do when you're setting up a new project for mixing in the Fairlight page is to define the audio tracks and busses you'll need to route and combine the elements of your mix.

This chapter covers how to create audio tracks and how to use busses to manage your mixes in the most efficient possible way. However, the Fairlight page gives you the flexibility to add or change your set up at any time; you can also concentrate on being creative and deal with any required housekeeping as your mix evolves.

Fairlight's FlexBus structure supports bus-to-bus, track-to-bus, or bus-to-track signal routing, along with expanded Dolby Atmos capabilities, including import, export, and manipulation of Atmos ADM files.

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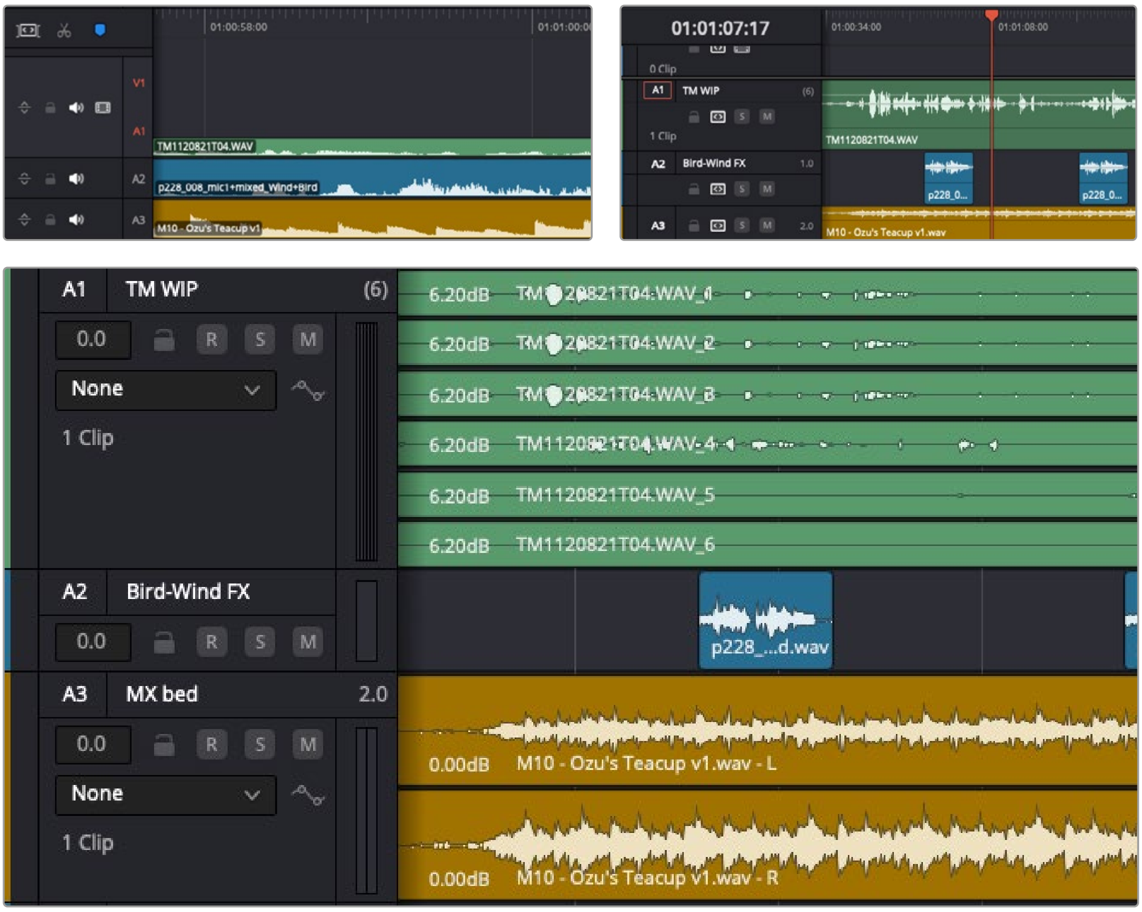
Audio Tracks

Each audio track in a DaVinci Resolve timeline corresponds to a single channel strip on the Mixer's left side. Depending on how an audio track has been configured, each audio track is assigned a specific audio format, such as mono, stereo, 5.1 or 7.1 surround, or Dolby Atmos. Track routing allows multiple audio channels within the clips on a track to be correctly routed to the proper audio output for monitoring and rendering via the lanes that can be seen within each track on the Fairlight timeline.

Audio tracks in DaVinci Resolve often contain multiple channels from a given clip, but how those individual audio channels are displayed depends on the page:

- The Cut and Edit pages display a single clip "lane" per audio track in the Timeline, regardless of how many channels the clips on that track are representing. The waveform displayed is a composite, showing a mix of the various channels within the clip.
- This can make it easier to work with multichannel tracks without the distraction of viewing multiple channel lanes, which could make things look a lot "busier." However, the tradeoff is that detailed activity on any of the various channels on that track are not seen.
- The Fairlight page displays the same number of tracks as the Edit page, but each track on the Fairlight page is divided into lanes, which shows each individual channel of a clip's audio. This additional visual information can help with editing and mixing.



(Top) 6 channel and stereo audio in the Cut and Edit pages represented by single composite tracks. (Bottom) The same audio in the Fairlight page shows the six channel and stereo tracks with multiple lanes that correspond to each of the file types.

Now that you understand how tracks work on the Fairlight page, the next important concept you need to understand in order to unlock the power of the Fairlight page is FlexBus, which lets you combine multiple audio tracks in different ways.

What Is a Bus?

A bus is simply a common signal connection point in an audio mixer. Busses can be mono, stereo, or any larger format, like 5.1 or Dolby Atmos 9.1.6 (where 16 audio signals are used). Bussed connections are mixed together into a single signal that can be controlled via a single bus channel strip. For example, by default a single bus called "Bus 1" combines the levels of every clip edited onto every track of a timeline into the mixed signal that is output to your speakers or headphones.

You can use busses in creative ways to organize mixing of tracks in a timeline. For example, if you have five audio tracks that have all of the edited dialogue audio clips for a particular program, you can route the output of all five dialogue tracks to a dedicated submix bus. This allows the combined levels from all the contributing dialogue tracks to be processed, adjusted, and mixed at once using a single channel strip's controls.

You can use multiple busses to organize a mix, including routing submix busses into other "main busses." Individual tracks can be routed to submix busses, then multiple submixes can be routed to one or more "main output" busses. For example, you could have four submix busses, one for German dialogue, one for English dialogue, one for Music, and one for Effects. You could route the German, Music, and Effects submix busses to a Main 1 bus to output the German version of the program, and route the English, Music, and Effects submix busses to a Main 2 bus to output the English language version of the program.

Surround Panning and Bussing

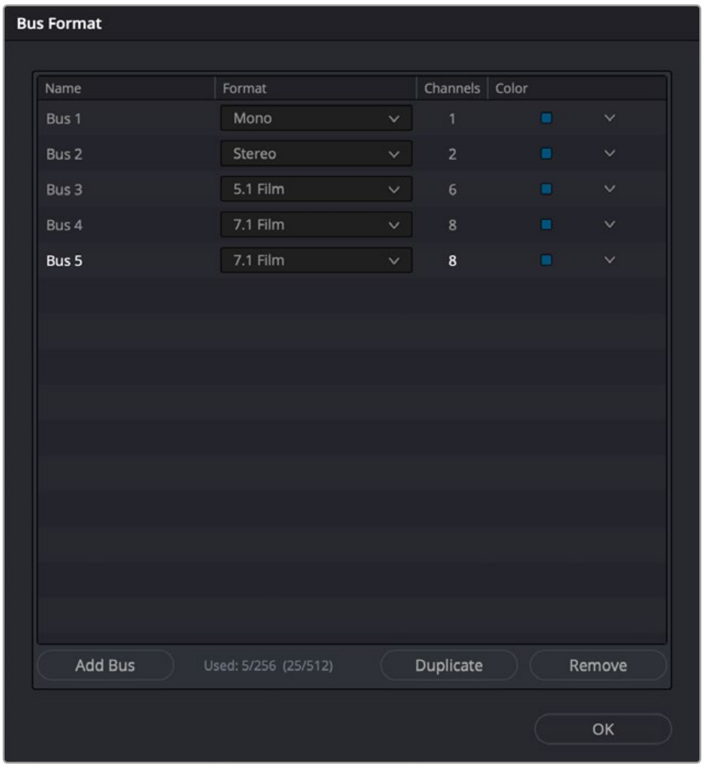
When working on surround or immersive formats, audio tracks from the Timeline are routed to busses via each channel strip's multi-format surround panner, so busses can be configured to accommodate specific audio formats, such as mono, stereo, LCRS, 5.1 surround, or 7.1 surround, and immersive formats, such as Atmos. By using multiple bus routing, you can even output different formats simultaneously.

Bus to Bus Routing and Mixing

Fairlight's FlexBus structure offers complete user flexibility for bus types and signal routing, allowing completely user-definable bussing, and making it possible to patch outputs and/or sends in any way you need, as dictated by your project. Each track can output to up to ten busses and sends with additional level and pan controls to a further ten busses. Busses can be sent to other busses up to six layers deep, facilitating complex stem building, processing, and allowing discrete deliverables.

User-definable busses allow for bus-to-bus, bus-to-track, or track-to-bus routing, with each bus having the ability to pass signals from mono to fully immersive formats, such as Dolby Atmos. As with any and all of the track types in Fairlight, these bus types can be changed at any time if needed.

The power of the FlexBus system is that it allows users to direct signals to many different places at one time, achieving complex mixing scenarios. Perhaps you need to generate two mixes that are identical in content but need to be of different output levels. You can designate two mix busses, one with an output level of -2dB true peak and one with an output level of -10dB true peak. The final mix signal is sent to one bus that is then broken into two more busses, one with a limiter set to -2dB and one set to -10 dB, creating these two different mixes at one time.



The FlexBus structure allows for many different bus track types to be created or changed.

Using Legacy Fixed Bussing

If you want to work using the previous method of Fixed Bus mapping, you can do so for new projects by opening the Fairlight panel of the Project Settings, and turning on the "Use fixed bus mapping" checkbox.

If your project has Fixed Bussing enabled and you want to change to FlexBus, then uncheck the "Use fixed bus mapping" checkbox. Note that once you have made the change it will not allow you to change it back to legacy bussing.

For more information *see Chapter 177, "Mixing in the Fairlight Page."*

Converting Older Fixed Bus Projects to FlexBus

Older Fixed Bus projects can be converted to FlexBus by doing the following:

- Open Project Settings > Fairlight.
- Under the Bussing heading, uncheck "Use fixed bus mapping."
- A dialog will appear allowing you to convert the project to FlexBus.

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Busses in Nested Timelines

When you nest a timeline inside another timeline that has busses set up for mixing in the Fairlight page, all bus routings continue to work as intended within the nested timeline, which exposes all channels via the default main bus (Bus 1) in the enclosing timeline. In this sense, the audio of the nested timeline can be considered to be a submix that outputs its resulting audio to the audio track it's edited onto. However, you can also decompose your nested timeline into its own bus structure within the main timeline you've imported into, exposing all of the original tracks as they were.

> This is a very powerful feature for combining work done at different times or by different contributors, *see Chapter 177, "Mixing in the Fairlight Page."* in section *Nested Audio Timelines* for more information.

Exposing Bus Tracks in the Timeline

You can expose any bus as a track in the Timeline. This makes it possible to view and edit automation that is applied to parameters on that bus.

To show a bus in the Timeline:

1. Enable the Toggle Automation button on the Fairlight toolbar. All busses are visible by default.
2. If you want to hide a bus, open the Index, and click the eye button for the bus you want to hide in the Timeline.
3. If you want to work with automaton on a bus, choose the desired automation curve you want to view from the dropdown menu in the track header controls.

Controlling Signal Flow

A good process for setting up editing and mixing in the Fairlight page is:

- Organize and configure the tracks on your timeline as required.
  For example, clips well organized on tracks, set track types, color, grouping, etc.
- Create the busses needed to organize the desired signal flow in the mix.
- Route the audio tracks, or any submix busses to the desired bus destinations for the mix layout.

Defining Audio Track Types

If you decide to create a new audio track, you have to choose what kind of audio track it will be. Right-clicking in the bottom audio portion of the Timeline track header reveals a contextual sub-menu that lets you create different kinds of audio tracks.

- **Mono:** Holds a single channel with only one lane.
- **Stereo:** Holds stereo left and right channels, with two lanes.
- **3.0:** Holds three channels, with three lanes in either LRC (Left, Right, Center) or LCR (Left, Center, Right) format.
- **4.0:** Holds four channels, with four lanes in either LRCS (Left, Right, Center, Surround), LCRS (Left, Center, Right, Surround), or Quad format.

- **5.x:** 5.0 holds five channels corresponding to a 5.0 surround mix, for a total of five lanes. For broadcast, SMPTE specifies Left, Right, Center, Surround Left, and Surround Right. For cinema distribution (5.0 Film), these tracks are ordered Left, Center, Right, Left Surround, and Right Surround.

  5.1 holds six channels corresponding to a 5.1 surround mix, for a total of six lanes. For broadcast, SMPTE specifies Left, Right, Center, LFE, Surround Left, and Surround Right. For cinema distribution (5.1 Film), these tracks are ordered Left, Center, Right, Left Surround, Right Surround, and LFE.

- **7.x:** 7.0 holds seven channels corresponding to a 7.0 surround mix, for a total of seven lanes. For broadcast, SMPTE specifies Left, Right, Center, Left Surround, Right Surround, Back Left Surround, and Back Right Surround. For cinema distribution (7.0 Film), these tracks are ordered Left, Center, Right, Left Surround, Right Surround, Back Surround Left, and Back Surround Right.

  7.1 holds eight channels corresponding to a 7.1 surround mix, for a total of eight lanes. For broadcast, SMPTE specifies Left, Right, Center, LFE, Left Surround, Right Surround, Back Left Surround, and Back Right Surround. For cinema distribution (7.1 Film), these tracks are ordered Left, Center, Right, Left Surround, Right Surround, Back Surround Left, Back Surround Right, and LFE.

- **Adaptive:** Holds up to 24 audio channels, each with its own lane within the track. An adaptive audio track can hold clips with different combinations of channels, up to the maximum number of channels allowed within that track. The number of channels allowable on a particular Adaptive track is user-definable (1–24) at the time that track is created. If you edit a clip with more channels into an Adaptive track that was created to hold fewer channels, the extra clip channels are muted.
- **Dolby Atmos:** There are several Dolby Atmos formats available: 5.1.2, 5.1.4, 7.1.2, 7.1.4, and 9.1.6. The naming of the channel configurations in the Dolby Atmos format includes the height channels in the nomenclature. Channel configurations are presented as three digits separated by periods, such as 7.1.4. The first digit describes the number of main, or ear-height monitoring channels that surround the listener. The second digit describes the number of subwoofer channels. The third digit describes the number of height channels, which are speakers positioned on, or in the case of a soundbar pointed to, the ceiling.

NOTE: The Dolby Atmos bus formats of 9.1.4, 9.16, and 22.2 are only available in DaVinci Resolve Studio and also require that Dolby Atmos be enabled in Preferences > Video and Audio I/O > Immersive Audio.

Adding Tracks (Contextual Menu)

There are two commands related to adding tracks in the right-click contextual menu on any audio track's header controls:

- **Add Track:** Adds a single audio track of the type you choose from a submenu.
- **Add Tracks:** Lets you insert as many tracks as you like, designating the track type and position in relation to other tracks in the Timeline.

Rearranging Tracks

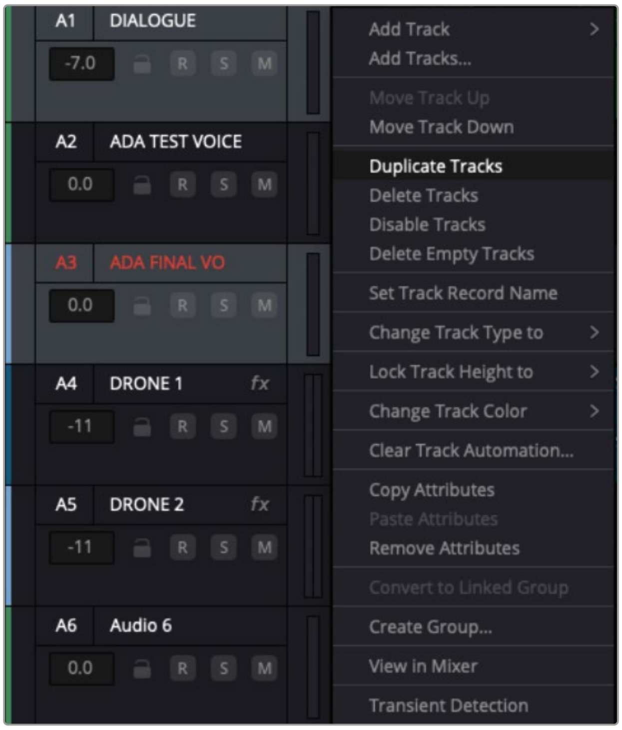
You can rearrange tracks by right-clicking in a track's header area and choosing either Move Track Up or Move Track Down in the contextual menu that appears. You can also move tracks in the Index by grabbing them and moving them to the desired position. This method works when moving multiple tracks at once.

Duplicating Tracks

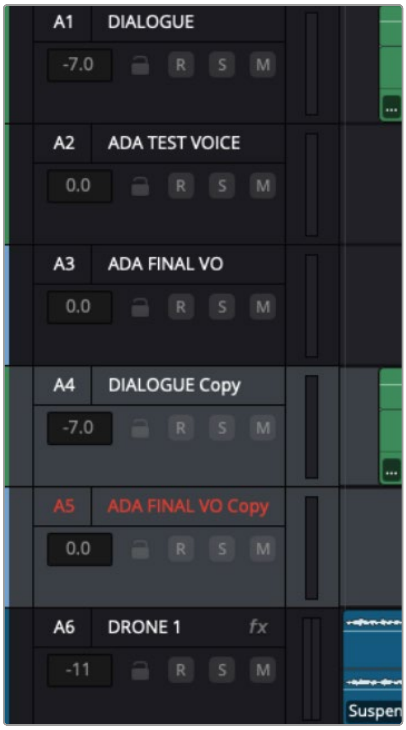
Tracks can be duplicated by right-clicking a track header and choosing "Duplicate Track" from the contextual menu.

You can duplicate multiple tracks simultaneously by selecting them, right-clicking any track header in the group, and choosing "Duplicate Tracks" from the contextual menu.

The names of duplicate tracks will be appended with the word "copy."



Timeline contextual menu - Duplicate Tracks



Duplicated tracks with appended names

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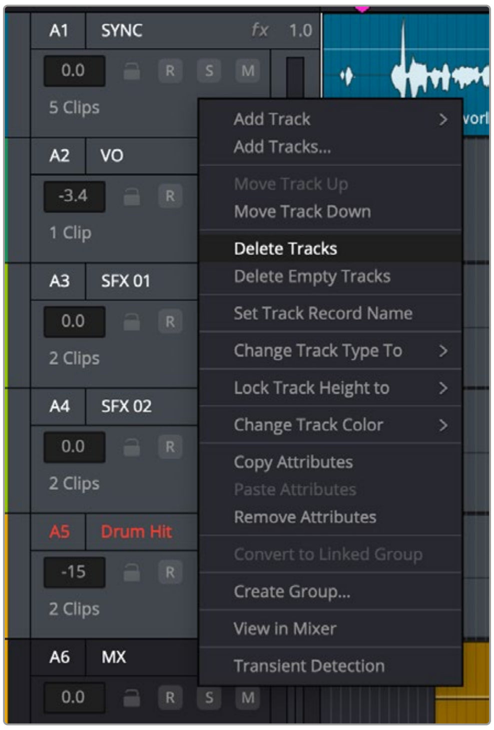
Deleting Tracks

Right-click within a track's Timeline header and choose Delete Track. If there are clips on a track you remove, they are also deleted from the Timeline, but preserved in the Media Pool.



Timeline contextual menu - Delete Tracks

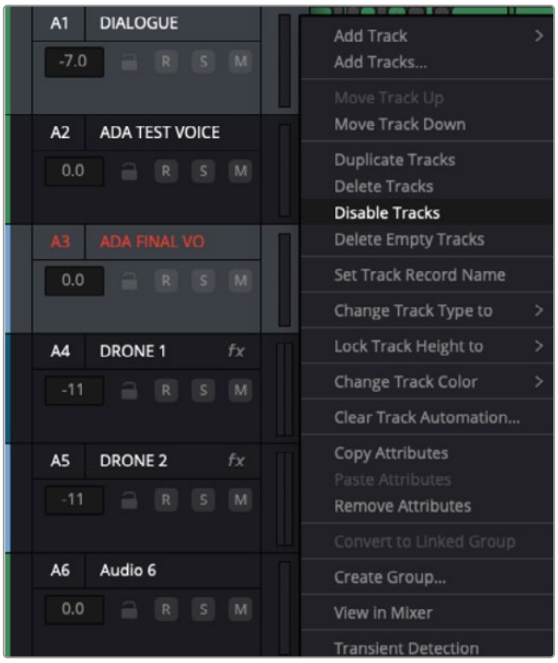
You can delete a selection of tracks in the Fairlight timeline by right-clicking any selected track header and selecting Delete Tracks from the contextual menu.

You can remove all empty audio tracks in the Fairlight timeline by right-clicking any track header and selecting Delete Empty Tracks from the contextual menu.

Disabling Tracks

You can disable an audio track by right-clicking its track header and choosing Disable Track from the contextual menu.

To disable multiple tracks simultaneously, select them, right-click any track header in the group, and choose Disable Tracks from the contextual menu.



Timeline contextual menu - Disable Tracks

Disabled tracks

To reenable your tracks, repeat one of the selection processes based on the number of tracks, and choose Disable Track or Disable Tracks from the contextual menu.

Changing Track Type

If you had set up your timeline with one kind of audio track, but you discover you actually need a different type, you can change it at any time. Just right-click anywhere in that audio track's Timeline header, and choose an option from the Change Track Type To submenu of the contextual menu.

Link Grouping

The Link Group function allows you to link mono tracks of material that are related and manipulate them as a one entity controlled by a single fader on a single channel strip. Only mono tracks can be used to create a link group; other track types such as stereo, 5.1, 7.1, Atmos, or Adaptive cannot be used.

Unlike a multi-channel track with lanes, a link group of mono tracks functions as independent, editable tracks in the Timeline. However, each track is mapped when choosing a track type, using one of the standard multi-channel mappings (stereo, 5.1, 7.1, Adaptive).

Link groups are extremely useful. For example, if you've been given independent ".L/.R" sides of a stereo mix, or a set of six independent related audio files that need to be assembled as a single 5.1 surround mix, or when you have surround channels that need to be specifically re-edited on a channel by channel basis.

To create a Link Group from individual mono tracks:

1. Create two or more Mono audio tracks that you want to group together. If you need to create a link group with a specific channel mapping, such as 5.1, make sure you create enough tracks (in this case, 6).

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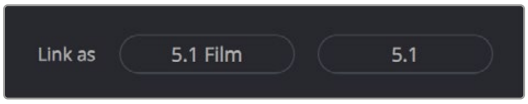
2 Choose Fairlight > Link Group.

3 When the Link Group dialog appears, mono audio tracks are represented by active buttons (all other track types are disabled, since they can't be linked). Click to enable the button of every track you want to include in the link group you're about to create. The available track type mappings when creating your group depends on how many tracks you've selected.

Selecting six tracks to use for creating a link group

4 After you've selected all the tracks you need, click one of the available "Link as" buttons below. In this example, six tracks have been selected, so you could click 5.1 Film or 5.1.



When you select enough tracks, you can create groups linked as specific surround mappings.

Afterwards, the tracks you selected should turn into a single block, showing they've been linked.

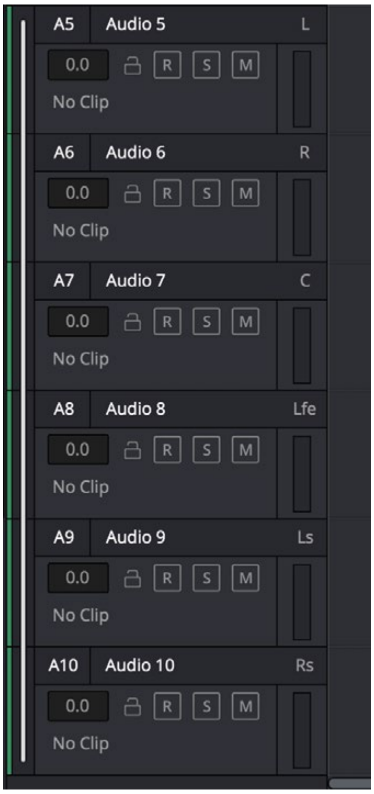
The Link Group window shows the link indicator line next to the track names.

Depending on the number of mono channels selected, Fairlight will offer possible linking options. For instance, when ten channels have been selected, both Atmos 7.1.2 and 5.1.4 are option choices.

The Link As option is dependent on the number of channels being linked.

5 Close the Link Group window when you're finished.

Once you've created a link group, you'll see a single bar on the left side of the headers, spanning the group. If the tracks are tall enough, each will be labeled, indicating its assigned channel based on the track mapping chosen in step 4. For example, with a 5.1 surround mapping, L, R, C, LFE, Ls, Rs, and so on. You also have the freedom to edit additional or different contributing elements of a surround mix into the appropriate track that represents that channel.



Tracks in a Link Group are labeled to identify which track corresponds to which surround channel.

Working Separately with Multichannel Audio File Elements

If you've edited a multichannel audio clip onto a multichannel track, you can convert that track and its contents into a Link Group of mono tracks, each of which contains a single clip for that track's channel. This can be useful if you need to fix a multi-channel surround audio clip with an incorrect track mapping. You can convert it into a Link Group, at which point you can easily rearrange the channels.

To create a Link Group from a single multichannel timeline:

Right-click the track header of a multichannel audio track, and choose Convert to Linked Group from the contextual menu. This automatically creates one new audio track for each channel, all of which are linked together. For example, converting a 5.1 audio track results in six new tracks with six individual audio clips (one for each channel), all of which are linked together.

If necessary, you can also unlink a linked group to turn it back into independent mono tracks.

To unlink a Link Group:

1. Choose Fairlight > Link Group.
2. When the Link Group dialog appears, select the link group you want to unlink.
3. Click Unlink.
4. Close the Link Group window when you're finished.

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SMPTE and Film order Standards

The variations of routing of the multichannel files are due to the path order of SMPTE or Film standards. They are:

5.1 film order: L, C, R, Ls, Rs, LFE

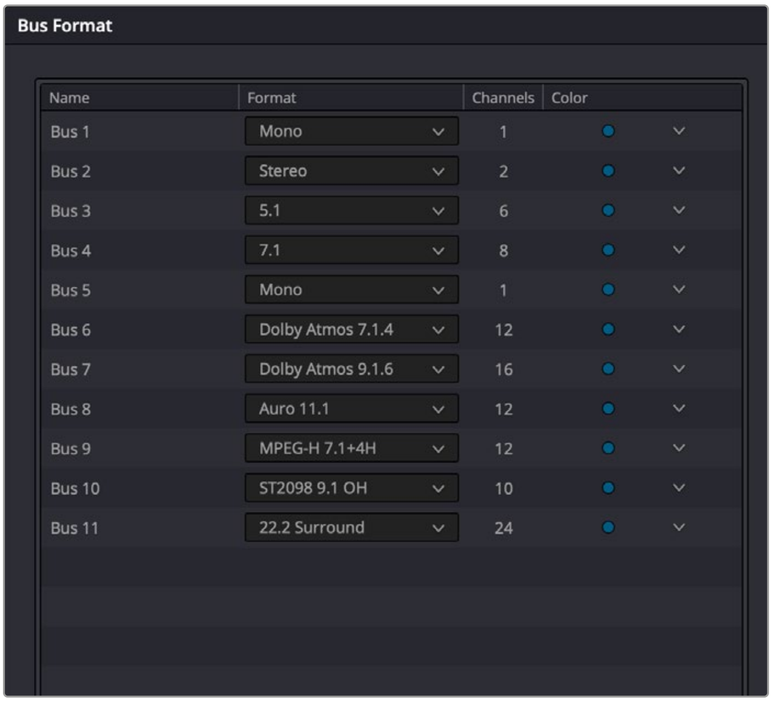
5.1 SMPTE order: L, R, C, LFE, Ls, Rs

7.1 film order: L, C, R, Lss, Rss, Lsr, Rsr, LFE

7.1 SMPTE order: L, R, C, LFE, Lss, Rss, Lsr, Rsr

Creating Busses

Choosing Fairlight > Bus Format opens the Bus Format window, which lets you create the busses you need (up to the limitations of your system) to organize the tracks and channels of your program.



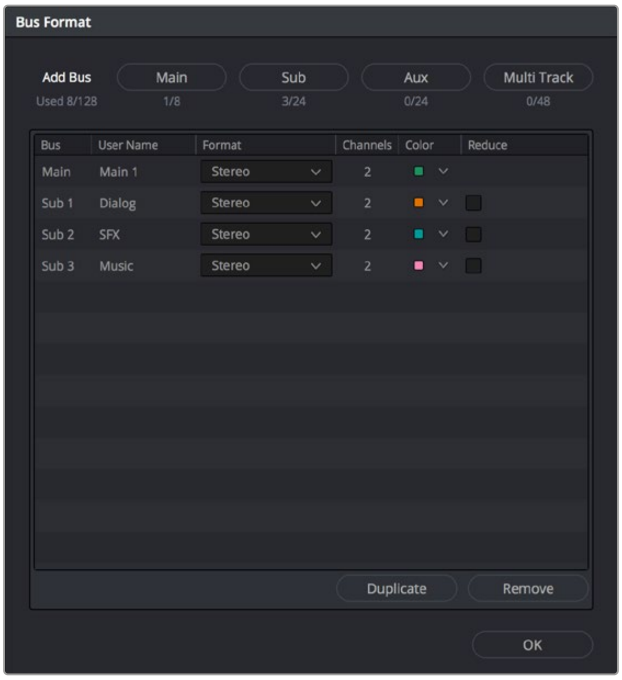
The depth of options in the FlexBus system

The Bus list lets you rename the bus, choose the format of each bus (a dropdown menu appears in the Format column of each entry of the list), shows the number of channels associated with a bus, and lets you color-code each bus (a Color dropdown lets you choose that bus's color). Simply click any item on the Bus list to select it, and choose different options from the Format and Color dropdown menus, or click on the Name of any bus to select it, and type a custom name.

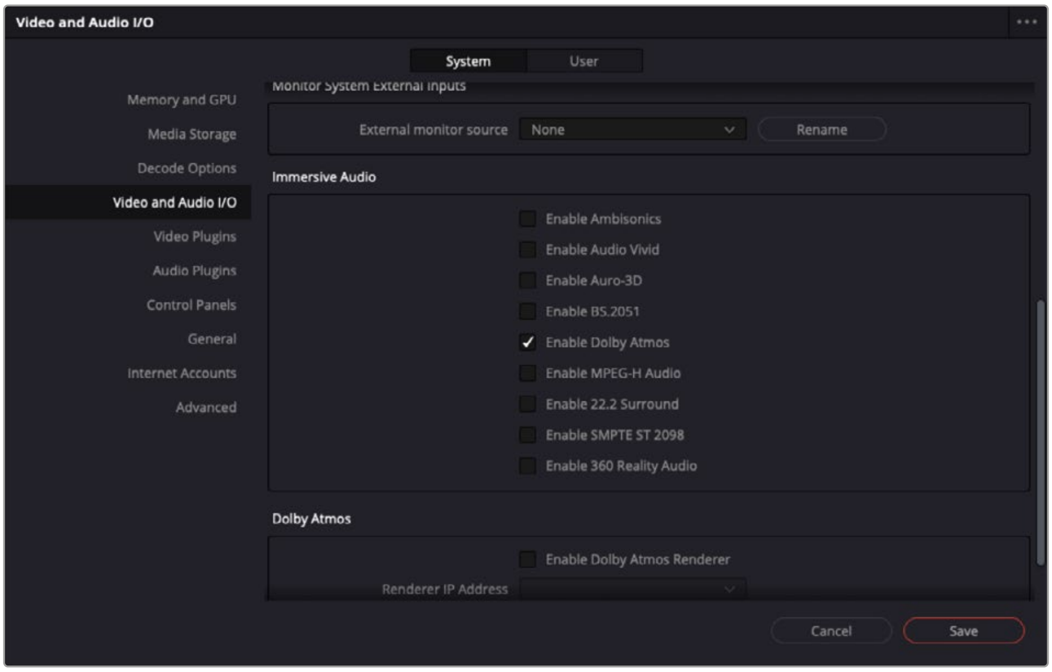
At the bottom of the list are three buttons that let you Add Bus, Duplicate, or Remove selected busses. When you're done modifying the available busses, you can click OK to accept the changes and close the Bus Format window, or Cancel to close the window without any changes (although all pre-existing busses remain in place). The bottom button row also has a Used tally of what has been used and what is available for your workstation.

Immersive Formats (Studio Version Only)

To turn on the immersive formats, go to Preferences > Video and Audio I/O > System Panel > Immersive Audio, and choose from among the appropriate options. Once enabled, the various immersive bus formats are available in the Bus Format panel’s Bus list.



The legacy Bus Format window lets you add busses to the mixer.



The Video and Audio I/O panel lets you enable or disable Immersive Audio options.

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Legacy Fixed Bussing

Working with the FlexBus topology is highly recommended for its flexibility. In addition, some newer features are only available using FlexBus. At some point, however, you may need to work with legacy Fixed Bussing in a new project. To use Fixed Bussing in a newly-created project, prior to adding any timeline, open Project Settings > Fairlight, and under the Bussing heading, check "Use fixed bus mapping." Now all bussing is handled with the legacy Fixed Bus topology.

Bus formatting with Fixed Bus

The legacy Bus Format window has four buttons that let you create the various Fixed Bus types. Creating a new bus, whether it's a Main, Sub, Aux, or Multi Track, adds the new bus to the list that appears beneath.

The legacy Bus list works the same as the FlexBus list with choices for rename, format, color code, and so on, and buttons that let you Duplicate or Remove selected busses.

When you're done modifying the available busses, you can click OK to accept the changes and close the Bus Format window, or Cancel to close the window (although any busses you've made remain in place). The bottom button row also has a Used tally of what has been used and what is available for your workstation.

Assigning Busses

Once you've created one or more busses, you can assign different tracks to specific busses or perhaps also busses to busses, busses to tracks, and the final Main bus destinations.

Bus Assignment Using the Mixer

You can easily assign your track(s) to any available busses simply by using the plus ("+") icon on a Mixer channel strip:

- Select a mixer channel, and click on the plus ("+") icon in the Bus Output or Bus Sends section.
- In the dropdown menu, choose the desired destination.
- A rectangle appears with the name of the bus, and the bus is now assigned.

Bus routing dropdown menu on a channel strip

TIP: You can quickly assign a bus to any selected group of tracks, or to all tracks, by holding down the Option key (Mac) or the Alt key (Windows) for all selected tracks or Command-Option (Mac) or Control-Alt (Windows) for all mixer channel strips prior to performing the operation. These shortcuts can save a lot of time in your workflow.

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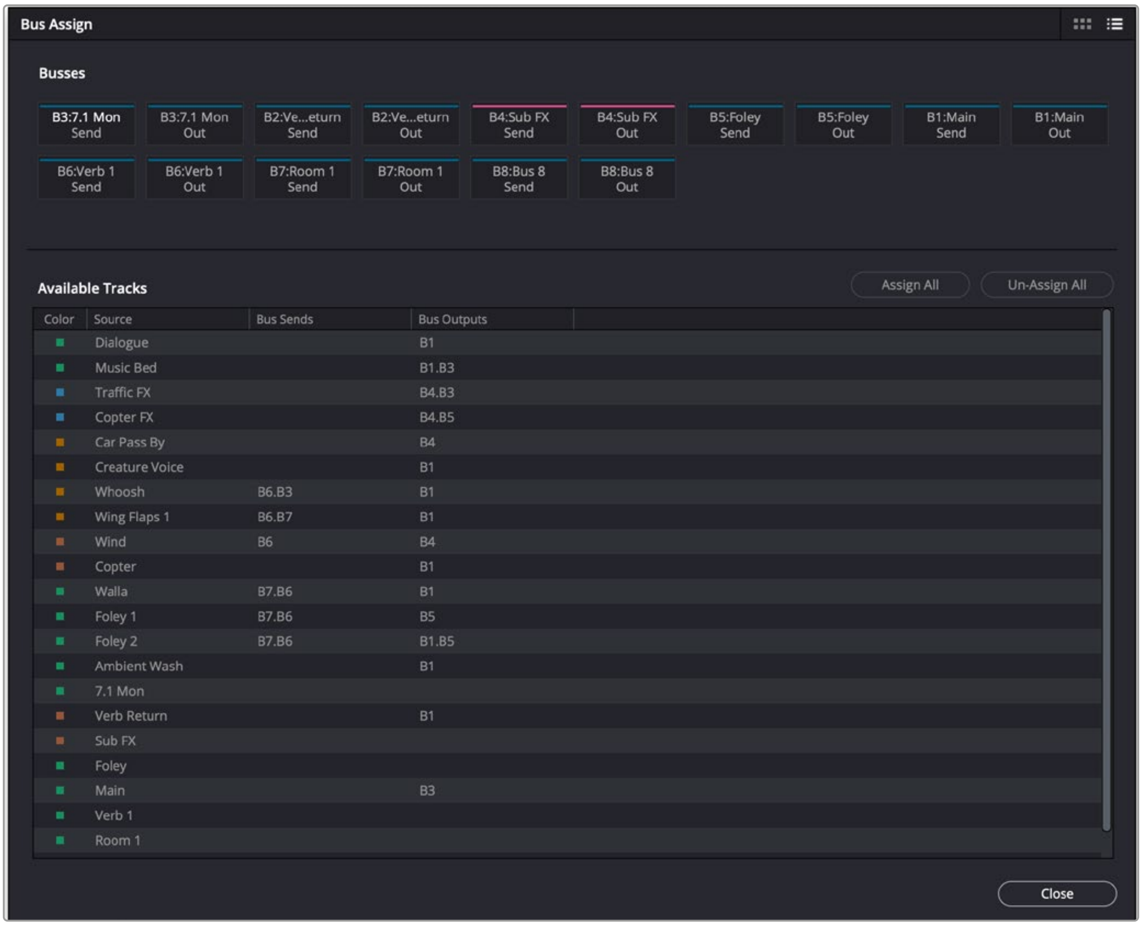
The Bus Assign Window

When you have a lot of busses or tracks to handle, the Bus Assign window lets you to easily manage connecting to all of them at once. Choose Fairlight > Bus Assign to open the window. Multiple bus assignments can be created within the dialog; these new assignments will be reflected in the Bus Outputs section on the channel strips in the Mixer.

The top shows the Send and the Out of each available bus, while the bottom shows a list of all available tracks and busses to connect to. The Bus Assign window defaults to List view, in which each bus and track is shown as a list, but by using the icon in the upper right of the window, it can be switched to Icon view, in which Available Tracks are shown as buttons.



Bus Assign window

Making Bus Assignments

Assigning Using List View

Click a button in the Busses section to select the Send or Out of that bus, and then click on any destination in the tracks/busses list below, or drag over a group of tracks or busses to assign to all of them. When assigned, the bus number will appear in the Bus Sends or Bus Outputs column.

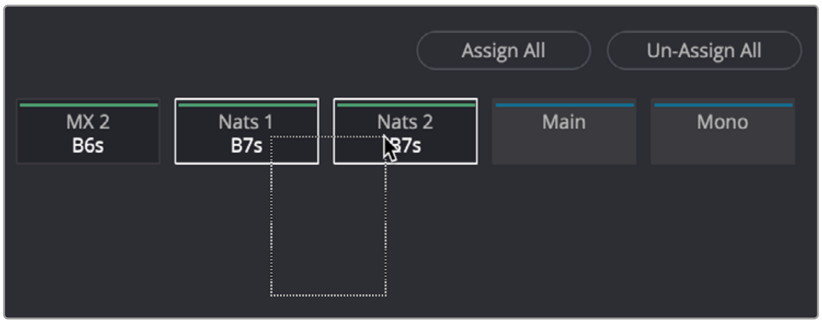
- **To assign every track, Sub, and Aux to a bus:** Click a button in the Busses section to select that bus, and then click Assign All.
- **To clear all track assignments from a particular bus:** Click a button in the Busses section to select that bus, and then click Unassign All.

  When you're done making bus assignments, click the Close button to close the dialog.

NOTE: While the dialog is open, you can undo any assignments you've made, one a time, or redo them, using Command-Z and Command Shift-Z respectively.

Assigning Using Icon View

- Click a button in the Busses section to select the Send or Out of that bus, and then either click on a target track button or drag a bounding box over all of the buttons for available tracks that you want to assign to that bus.
- Once assigned, the Available Tracks buttons display which bus they've been assigned to. When assigned, the Bus number will be followed by an "o" or an "s" to indicate if it's the send or the out of that bus.



Assigning multiple tracks to a bus in Icon view by dragging a bounding box

Setting Signal Paths

While bus creation and assignment allows you to submix tracks or route from one bus to another, you often need to route signals from individual, external sources to a track or bus. For example, if you need to record audio to a track, you need to patch the audio input on your hardware to the track you want to record to. Making this type of connection is called "patching" in Fairlight and is accomplished using the Patch Input/Output window, which is available on the Fairlight, Edit and Deliver pages, providing patching changes on any of these.

Using the Patch Input/Output Window

Choosing Fairlight > Patch Input/Output opens the Patch Input/Output window, which can be displayed in either List (default) or Icon view. This window is split into two halves, with the left half containing whichever Source controls you choose and the right half containing whichever Destinations you choose.

Creating a Patch

By default, the Patch Input/Output window shows the available Audio Inputs as the Source and the Track Inputs as the destination. This makes it easy to patch whatever audio source (such as a microphone connected to a USB audio interface) to a specific audio track of the Timeline to prepare for recording. Patching and unpatching a source to a destination is straightforward. In the following screenshot, the Audio 1 Input from an audio interface is highlighted and is being patched to a track named "Walla."

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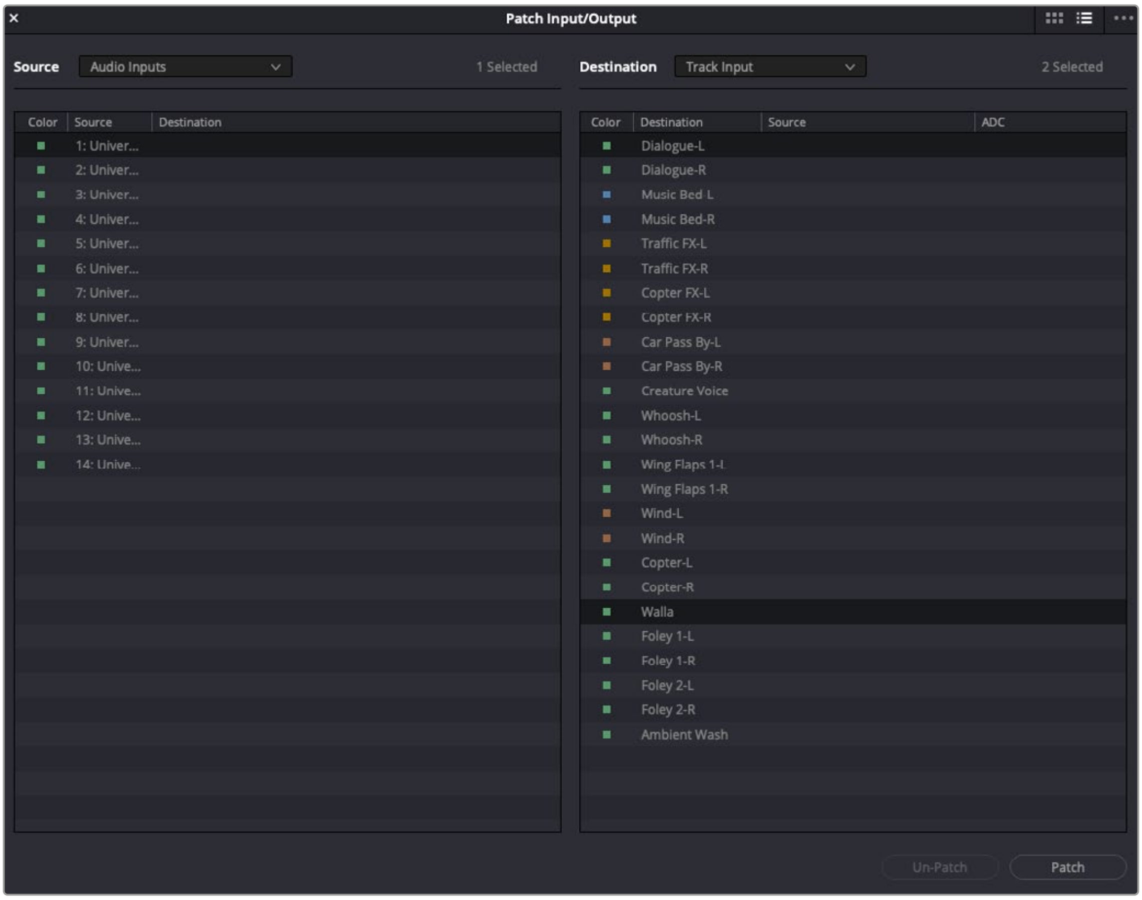

The Patch Input/Output window showing Source Audio Inputs and Destination Track Inputs

Patching a source to a destination:

1. Choose a type of source from the Source dropdown menu at the upper left-hand side of the window.
2. Click the item on the source list (or action button in Icon view) that you want to patch from on the left side of the Patch Input/Output window.
3. Choose a destination from the Destination dropdown menu at the upper right-hand side of the window.
4. Click the list item (or action button) of the destination you want to patch to on the right side.
5. Click the Patch button at the bottom right of the window. The source and destination will both display the connection they're patched to.

To unpatch a source and destination pair:

1. Click a list item or action button corresponding to a source or destination you want to unpatch.
2. Click Unpatch.

Patching in Icon view

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Choosing Source and Destination Controls

The Audio Source and Destination dropdown menus let you choose different categories of sources and destinations to patch together.

The following Source options are available:

- **Audio Inputs:** The available physical audio inputs on your workstation, for example the Fairlight SX-36 audio interface, MADI, third party interfaces (for example, USB or Thunderbolt), or system audio. Useful when patching to record audio.
- **Bus Out:** Any Busses.
- **Monitor Direct:** Monitor system Direct Out, routed post fold up/down matrix, pre the Monitor VolumeLevel/Dim/Mute.
- **Monitor Out:** Monitor system output. Post fold up/down matrix and Monitor VolumeLevel/Dim/Mute.
- **System Generator:** Provides various utility sources, including beeps and timecode generation:

  OSC: Provides a test oscillator. The oscillator controls appear in a floating window accessed via Fairlight > Test Tone Settings.

  Noise: Uses the Noise generator with choice of white or pink noise, which can be selected via Fairlight > Test Tone Settings.

  Beeps: Refers to beeps generated by operations on the ADR panel, with controls found on the ADR setup page.

  Timecode generator: Generates timecode when enabled and the transport runs, with controls provided in a floating window accessed via Fairlight > Remote Control Settings.

- **Track Direct:** Track Direct Out, can be pre or post the track fader, with an offset.
- **Track Reproduction:** This is the track playback signal, directly from disk, before any processing.
- **Dolby Atmos Renderer:** The output from the Dolby Atmos Renderer in the currently selected Atmos output monitoring format. There is also a parallel stereo binaural output to feed headphones.

The following Destination options are available:

- **Audio Outputs:** For example, the Fairlight SX-36 audio interface, MADI, third party interfaces (for example, USB or Thunderbolt), or system audio. Useful when patching to playback audio to speakers or headphone systems.
- **Talk Back:** The system for patching the General Purpose Input and General Purpose Output used for talkback.
- **Track Input:** The inputs to the available audio tracks in the current timeline, which are the inputs to the Record and Thru path.
- **Dolby Atmos Send:** You will need to patch this manually if you are creating original content. If you import a Dolby Atmos master file, the Send patching for the bed and object tracks will be created automatically. When an external Dolby RMU renderer is enabled, all sources patched to these sends will be mirrored on the physical outputs to the external renderer, as defined by the base audio outputs set in system preferences.

Legacy Fixed Bus Patching

If you choose to use the legacy Fixed Bussing in the Project Settings, the options appear a bit differently.

The following Audio Source options are available:

- **Audio Inputs:** The available physical audio inputs on your workstation, for example SX-36, MADI, or system audio. Useful when patching to record audio.
- **Monitor Direct:** Monitor system Direct Out. Post fold up/down matrix, pre the Monitor VolumeLevel/Dim/Mute.
- **Monitor Out:** Monitor system output. Post fold up/down matrix and Monitor VolumeLevel/Dim/Mute.
- **Main Direct:** Main Bus Direct Out; can be pre or post the Main bus master fader, with an offset.
- **Main Out:** Main Bus Out; always post the Main bus master fader.
- **Main Send:** Main bus master insert send.
- **System Generator:** Provides various utility sources, including beeps and timecode generation:

  OSC: Provides a test oscillator. The oscillator controls appear in a floating window accessed via Fairlight > Test Tone Settings.

  Noise: Uses the Noise generator with choice of white or pink noise, which can be selected via Fairlight > Test Tone Settings.

  Beeps: Refers to beeps generated by operations on the ADR panel, with controls found on the ADR setup page.

  Timecode generator: Generates timecode when enabled and the transport runs, with controls provided in a floating window accessed via Fairlight > Remote Control Settings.
- **Track Direct:** Track Direct Out; can be pre or post the track fader, with an offset.
- **Track Reproduction:** This is the signal from the track playback, before any processing.
- **Track Send:** Track Insert send.

The following Audio Destination options are available:

- **Audio Outputs:** The available physical audio outputs on your workstation. For example, the Fairlight SX-36 audio interface, MADI, third party interfaces (for example, USB or Thunderbolt), or system audio. Useful when patching to play back audio to speakers or headphone systems.
- **Main Return:** Main Bus master insert return.
- **Talk Back:** The system for patching the General Purpose Input and General Purpose Output for talkback.
- **Track Input:** The inputs to the available audio tracks in the current timeline, which are the inputs to the Record and Thru path.
- **Track Return:** Track insert return.
- **Dolby Atmos Send:** You will need to patch this manually if you are creating original content. If you import a Dolby Atmos master file, the Send patching for the bed and object tracks will be created automatically. When an external Dolby RMU renderer is enabled, all sources patched to these sends will be mirrored on the physical outputs to the external renderer, as defined by the base audio outputs set in system preferences.

Using a Channel Strip's Input Menu

The Input dropdown menu at the top of each track's channel strip in the mixer provides some shortcuts for patching different inputs and busses to the tracks of your mix. Each option in this menu makes the Patch Input/Output window appear with various Source and Destination selections automatically set up.

Input

The Patch Input/Output window appears set up to let you patch different inputs (such as the system audio input) to the tracks of the timeline. This makes it fast for setting up audio inputs in preparation for recording.

Bus

A shortcut to open the Patch Input/Output window (discussed previously in this chapter) that lets you patch Bus Out or Bus Sends to Timeline track channels.

Path Settings

This opens the Path Settings window for the track, which contains controls for adjusting the input level of audio signals coming from an input/output device.

The Path Settings window showing audio inputs and track inputs

These parameters are as follows:

Mic/Inst

Controls will only appear on this panel if you have connected channels 1 or 2 of a Fairlight SX36 audio interface to your system. If connected, you can remotely control all of the options (including level) for the mic/instrument inputs of the SX36 if they are assigned to the channel. If there is no connection to an SX36, the area is empty.

Record Level

- **Record:** This button is linked and identical to the Record Enable button on the channel strip; here for convenience. If you hit one, it will enable the other.
- **Thru:** Lets the input signal to pass into the mixer without enabling a record path. This is ideal when you want a source signal to always be available and just want to monitor it.
- **Record Level:** Allows you to apply a digital gain adjustment to the record path to disk, post the output of your audio interface's analog-to-digital converter.

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TIP: The mixer channel in question can also be switched to Thru mode by Command-clicking the Track Arm button, in either the Mixer or Track Header. The button will turn green and 'T' will be displayed within it.

Normally, this control should be left at 0.0 (no change, unity gain), as it affects the level you are recording to disk. It is best practice to use the level controls on your audio interface to control the input level into DaVinci Resolve in order to maximize audio fidelity. However, there might be a time where you need a bit more level, or may not have access to an audio interface's controls, and in those cases you can adjust the input.

Trim

- **Polarity:** This button inverts the polarity of the signal coming into the channel strip (sometimes referred to as "flipping the phase").

  For example, you may have an input signal, like an explosion, where the transient attack of the signal produces with a massively positive-going waveform (where the waveform mostly appears above the zero line).

  If you invert the polarity, the signal will now be mainly negative-going and the waveform will be concentrated beneath the zero line. Inverting polarity is sometimes used to more closely align signals from multiple microphones and can be used creatively to affect frequency response of such a signal.

- **Trim Level:** The Trim knob lets you adjust the level of the signal coming "off of disk" during playback to optimize the level feeding effects and the busses. This is helpful when you want to slightly trim an otherwise perfect element up or down in your mix.

  Trim adjustments are applied post the recorded signal and Track FX (which processes directly "off of disk"), and pre all other effects, and do not affect the recording level.

Direct Output

Each audio channel strip can enable a direct output that can be used to feed any other input destination. You can patch this source using the Track Direct choice in the Patch Input/Output dialog via the Source dropdown.

- **On/Off:** Enables and disables the direct output.
- **Pre:** Sets the direct output tap-off point to be pre (before) the channel fader. On by default.
- **Level Control:** Sets the direct output level from minus infinity (fully off) to +10 dB.
  Default is 0.0 (unity gain).

For more information *see Chapter 171, "Setting Up Tracks, Busses, and Patching."*

Insert

This button is linked and identical to the Effects In button on the channel strip; here for convenience. Clicking this button in one location enables the other and switches all Fairlight FX, AU, or VST channel effects in or out of the signal path.

Input Menu with Legacy Fixed Bussing

When using the legacy Fixed Busing, the options there are a bit different. Here is how they look when that is enabled.

Input

The Patch Input/Output window appears set up to let you patch different inputs (such as the system audio input) to the tracks of the Timeline. This makes it fast for setting up audio inputs in preparation for recording.

Aux Bus

A shortcut to open the Patch Input/Output window (discussed previously in this chapter), which appears set up to let you patch different Aux busses to specific submix and Timeline track channels.

Sub Bus

A shortcut to open the Patch Input/Output window (discussed previously in this chapter), which appears set up to let you patch different Sub (submix) bus channels to specific Timeline track channels.

Main Bus

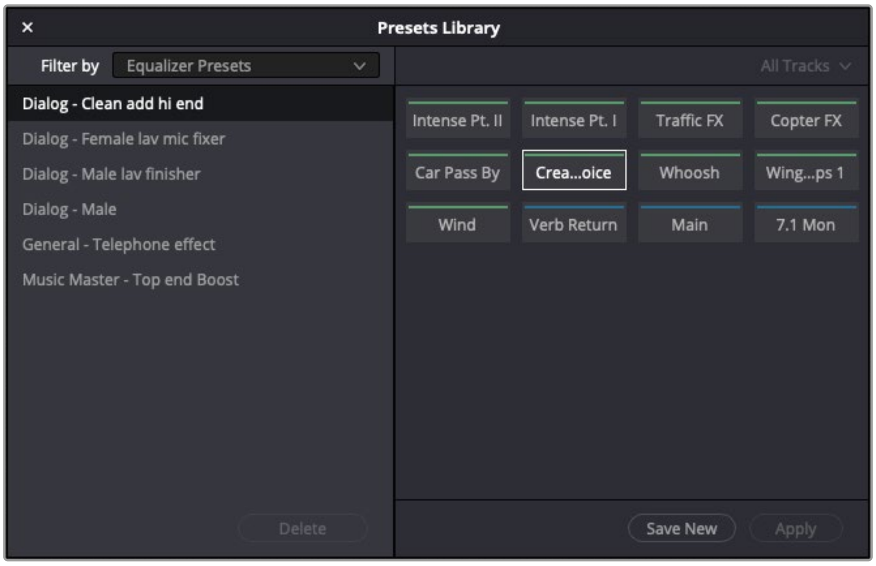
A shortcut to open the Patch Input/Output window (discussed previously in this chapter), which appears set up to let you patch different Main bus channels to specific Timeline track channels.

Path Settings

This option opens the Path Settings window, which contains controls for adjusting the input level of audio signals coming from an input/output device.

The Fairlight Presets Library

The Fairlight page offers a powerful and flexible preset system that allows you to store and recall presets for everything from the complete configuration of the audio page to individual presets for plugins. To access the presets library, choose Fairlight > Presets Library.



Fairlight Presets Library with stored presets on the left panel and destination tracks on the right

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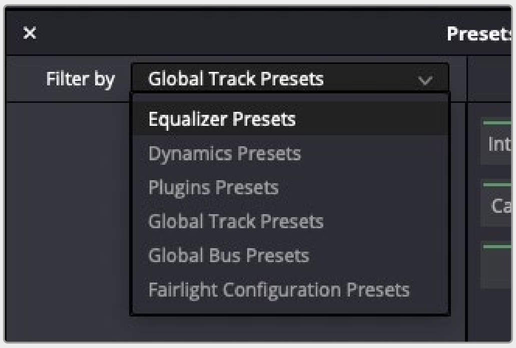
A Filter By dropdown menu on the upper left lets you choose the type of preset you want to work with.



Filter By menu of preset types

The choices are:

- **Equalizer Presets:** Presets for the built-in channel EQ.
- **Dynamics Presets:** Presets for the built-in Dynamics Processor.
- **Plugins Presets:** Fairlight FX, AU, or VST.
- **Global Track Presets:** All settings for a Mixer channel strip.
- **Global Bus Presets:** All settings for a Mixer bus.
- **Fairlight Configuration Presets:** Many parameters of the Fairlight page are stored as a preset including:
  - Track height.
  - What tracks are shown or hidden in the Tracks pane of the Index.
  - Split point in the Mixer.
  - Full Track vs. Small Track views in the Mixer.
  - Track Groups enable status.

NOTE: Some items are not currently stored in Fairlight Configurations, such as which panels are shown (for example, Media Pool or Index), Meter Panel show/hide, Fairlight video window docking, and size if the video window is set to be floating.

Using the Presets Library

Select the preset type you want to work with from the Filter By dropdown menu on the left.

The tracks list on the right (which appears for most preset types) acts as both a list of sources that contain information you want to store and a list of destinations that you want to assign to when you have presets defined. To store and recall and assign presets:

- If you're storing a preset for the first time, select the source track in the tracks list on the right of the window that you'll be grabbing the preset type from (unless it's a Fairlight Configuration preset, which has no tracks to chose from, as it is global).
- Click the Save New button to save your preset.

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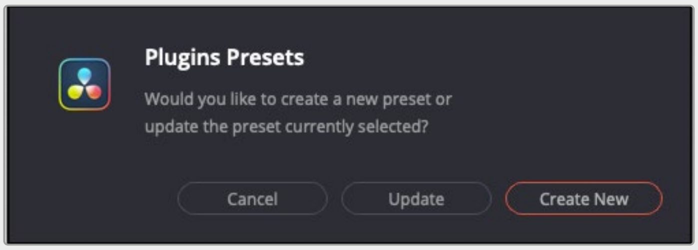
- To load your preset onto another track, deselect any tracks currently selected in the track list on the right, and select the track(s) you want to assign to. Click Apply and the preset is loaded.
- To save a new version of an existing preset or update your present version, first choose a track in the track list that has your preset assigned to it. Then click the Save New button, and a dialog appears that allows you to choose to update the current version of the preset or save a new preset based on the current settings.
- To Delete a preset, select it in the list, and click the Delete button.



Update or Create New preset dialog

FAIRLIGHT